Lab 5: Static and Default Route Configuration

Objectives

- To enable communication between different local area networks (LANs) by manually configuring routes across an internetwork consisting of two routers.
- To observe how the router processes a static route by looking up the next-hop IP address in the routing table.

Theory

Routing refers to the process of directing data packets from a source device to a destination device across a network. Routers function similarly to postal services: they receive packets, analyze the destination IP address, and consult their routing tables to determine the most appropriate path for forwarding.

Static Routing

Static routing is a routing technique where the network administrator manually specifies routes in a router's routing table. Unlike dynamic routing, static routes do not update automatically when changes occur in the network topology.

Routing Command

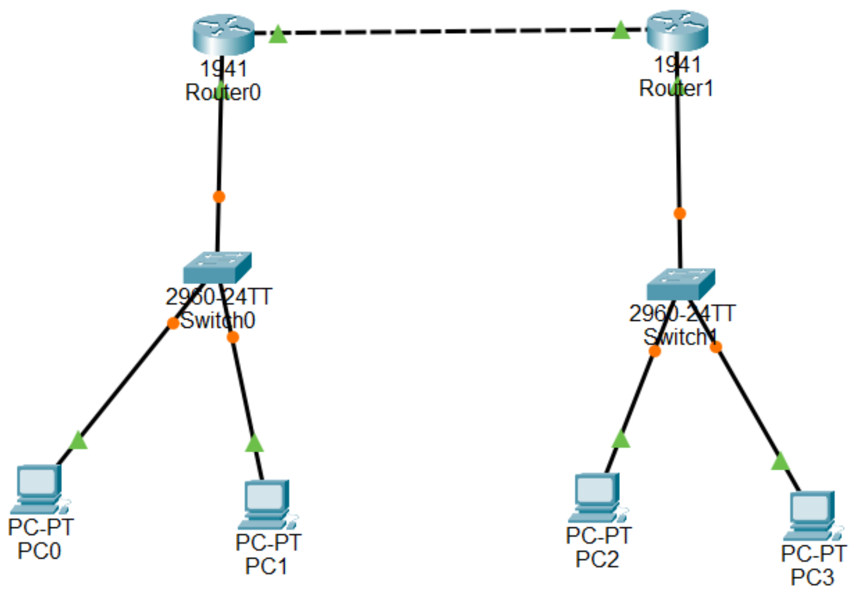
On Cisco routers, static routes are configured using the following command:

ip route <destination network> <subnet mask> <next hop IP>

Key Features of Static Routing

- Efficient resource usage, as no routing updates are exchanged between routers.
- Enhanced security since routes are not shared dynamically, reducing the risk of malicious route manipulation.

Network Configuration



PCs were configured with appropriate IP addresses, subnet masks, and default gateways to match their respective LANs. Routers were assigned interface IP addresses and static routes to enable communication between the networks.

For PCs:

Device	IPv4 address	Subnet Mask	Default Gateway
PC0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	192.168.1.3	255.255.255.0	192.168.1.1
PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

For Routers:

Device	IPv4 address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
Router 1	192.168.2.0	255.255.255.0	10.0.0.2	gig 0/0 gig 0/1	10.0.0.1 192.168.1.1
Router 2	192.168.1.0	255.255.255.0	10.0.0.1	gig 0/0 gig 0/1	10.0.0.2 192.168.2.1

Observation:

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time<1ms TTL=126
Reply from 192.168.2.2: bytes=32 time=1ms TTL=126
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Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Default Routing

Default routing is a routing technique where a router forwards all packets with unknown destinations to a single predefined route, usually called the default gateway. This method is often used in small networks or at the edge of a network to simplify routing, as it eliminates the need to maintain multiple specific routes.

Routing Command

On Cisco routers, a default route is configured using the following command:

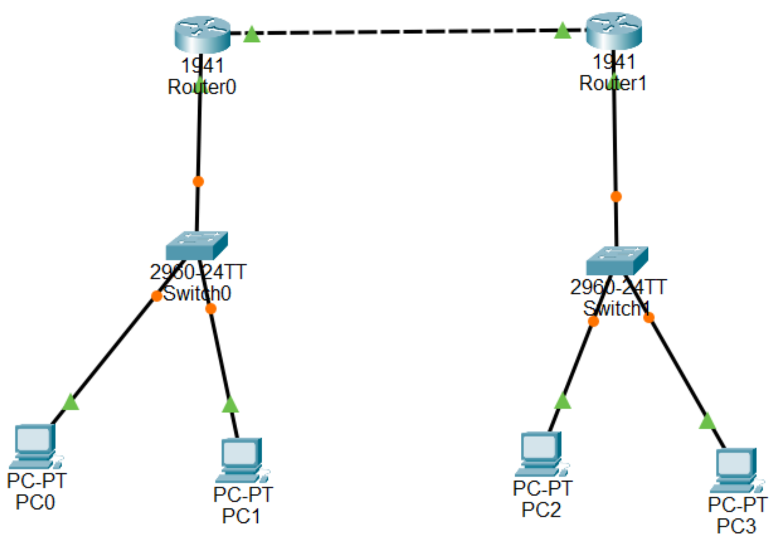
ip route 0.0.0.0 0.0.0.0 <next hop IP>

Here, 0.0.0.0 0.0.0.0 represents all possible networks, and <next hop IP> is the IP address of the gateway router to which unknown packets should be forwarded.

Key Features of Default Routing

- **Simplified configuration**: Reduces the complexity of the routing table, especially in networks with a single exit point.
- **Efficient for small networks**: Useful where maintaining detailed routes is unnecessary.
- **Acts as a fallback route**: Ensures packets are still delivered even if there is no specific route in the routing table.

Network Configuration



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PC2	192.168.2.2	255.255.255.0	192.168.2.1
PC3	192.168.2.3	255.255.255.0	192.168.2.1

For Routers:

Device	IPv4 address	Subnet Mask	Next Hop	Ethernet Cable	Default Gateway
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Results

After configuring static routes on both routers, successful communication was established between the two LANs. Connectivity was verified through ping tests between hosts on different networks, with only minor initial packet loss.

Similarly, configuring default routes on both routers allowed traffic destined for unknown networks to be forwarded correctly. End-to-end connectivity was confirmed through successful ping operations.

Discussion

The lab demonstrated that static routing effectively enables communication between separate LANs when routes are manually defined. It also highlighted the advantage of reduced bandwidth usage since no routing updates are exchanged.Default routing provided a simpler and more scalable solution for handling unknown destinations. Although some initial packet loss was observed, stable communication was achieved afterward.

Conclusion

This experiment successfully illustrated the configuration and functionality of static and default routing using Cisco Packet Tracer, reinforcing the practical understanding of routing concepts.